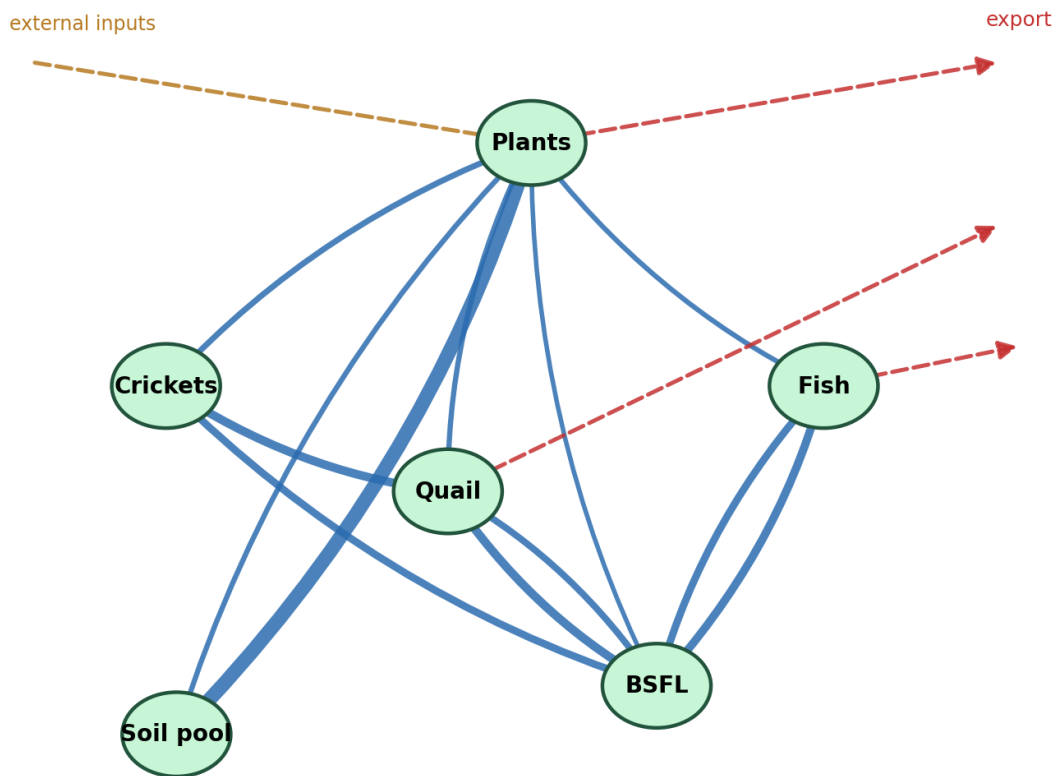


# THE LOOP

## A Practical Research Report

*Turning waste into food, food into jobs, and jobs into self-sustaining communities.*



The production loop: plants feed insects and fish; insects feed birds and fish; every waste stream returns as feed or fertilizer.

**Prepared by** [Your name / organization]

**Date** September 2026

**Audience** Community members, buyers, funders, residents

**Status** Pre-construction — design validated by quantitative modeling

# 1. Executive summary

This report summarizes, in plain language, the complete design of a community-owned circular protein farm: what it is, why the science says it works, what it costs, when it pays for itself, and how to be part of it.

- **The problem.** Tens of millions of Americans are food-insecure while 30–40% of food is wasted. Hunger in the United States is a price and logistics failure, not a shortage of calories.
- **The idea.** A closed production loop — plants feed crickets, quail, and fish; their "wastes" feed black soldier fly larvae; the larvae come back as premium feed; frass returns to the soil. Six sellable product lines come out of one nutrient circle.
- **The validation.** Every claim in this report is backed by a quantitative model published as a preprint (arXiv) and a full executable simulation. Nothing here is "we think" — it is "we computed."
- **The economics.** A two-cell farm (serving ~20 residents) builds for \$13k grid-powered or \$37k fully off-grid, earns roughly \$4,300/month at steady state, and reaches cash break-even in about 18–26 months depending on energy choices.
- **The financing.** Buyers pre-purchase future production (food futures) to fund construction; USDA grants (LFPP, VAPG, BFRDP, GusNIP) provide matching non-dilutive capital; investors can take revenue-share notes.
- **The ask.** \$150,000 builds the first farm: production, housing seed, and twelve months of working capital.

**Bottom line:** a farm that feeds its community first, sells the surplus to pay its bills and its investors, wastes nothing, and can be copied by any community with land and willing hands.

## 2. The problem: hunger is a price problem

Roughly 47 million Americans lived in food-insecure households in 2023 (USDA) — in the world's richest food economy. The pattern is consistent: households do not run out of calories first; they run out of **nutritious** calories first. Fresh protein and produce are the most expensive calories per nutrient, so diets degrade exactly where health is already most fragile.

At the same time, an estimated 30–40% of the food supply is wasted — thrown away as manure, frass, sludge, and residue. The calories exist. They are simply classified as waste and routed to disposal instead of to dinner.

Finally, the work of producing food is usually located far from where people are hungry and unemployed at the same time. This project fixes that geographic mismatch: put the farm where the need is, house the workers beside it, and sell the surplus locally.

### 3. The idea: one loop, six product lines

The farm is a nutrient circle, not a collection of separate enterprises. Plants partition their output among crickets, quail, fish, and black soldier fly larvae (BSFL). Crickets feed the quail. Quail and fish send their wastes to the BSFL farm. The larvae — a premium protein and fat source — return as feed for quail and fish, and the frass returns to the soil as fertilizer. External inputs are limited to seed, minerals, water, sunlight, and labor.

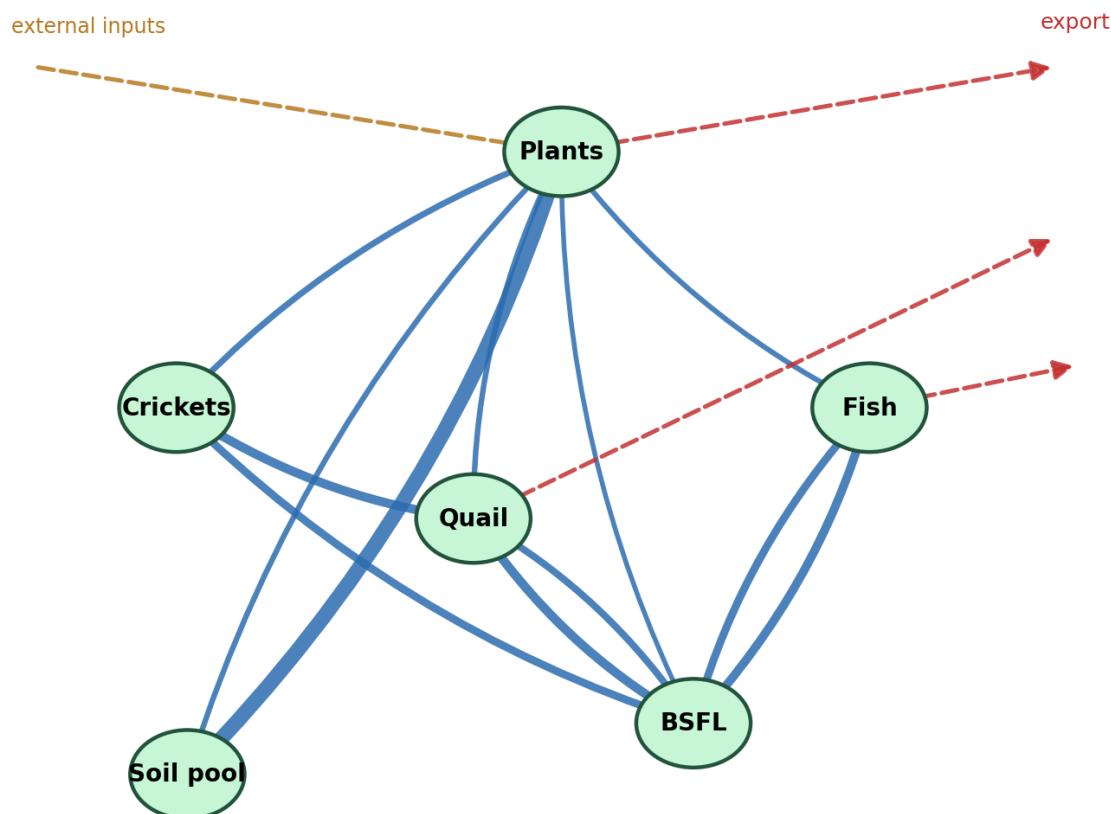


Figure 1 — the production network. Solid arrows are measured nutrient flows; dashed arrows are external inputs and export.

Weekly outputs: eggs, quail meat, tilapia, BSFL protein meal, frozen crickets, fresh sprouts and greens, and compost. Because feed is largely recovered rather than purchased, production cost sits structurally below conventional producers — and that margin is what makes nutritious food affordable.

## 4. The science, in plain terms: we modeled it before building it

**Waste is food with a measurable price tag.** Manure, frass, and sludge carry real nutrition. In our model, cricket frass alone scores 1.0 on the protein requirement of black soldier fly larvae — the "waste" is a complete feed ingredient, not a disposal problem. Each waste stream is scored against the animal that will eat it.

**Most of the food comes from the loop itself.** When the model runs the full circle, 34–60% of steady-state protein output arrives through indirect recycling pathways — flows that a conventional accounting of "inputs minus waste" would miss entirely. The loop is not a sustainability garnish; it is the majority of production.

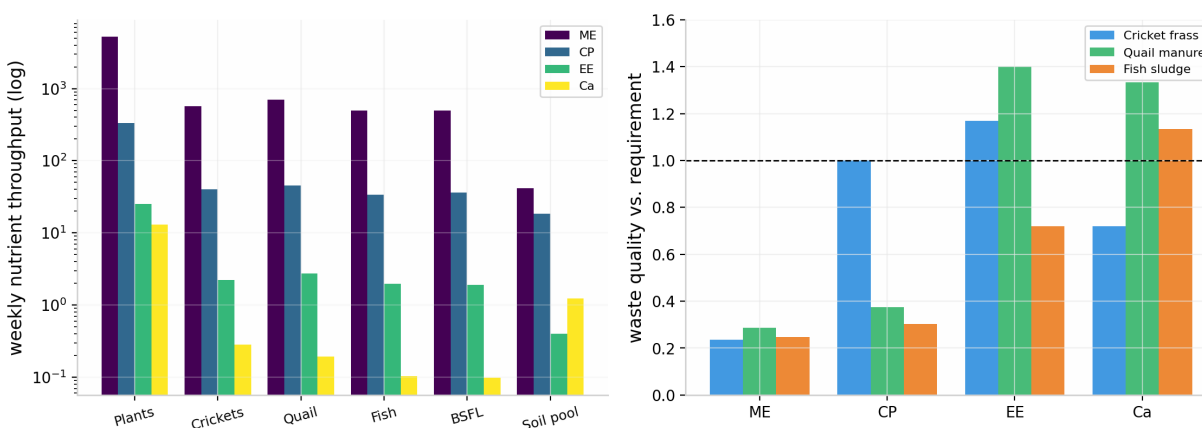


Figure 2 — (a) steady-state weekly nutrient throughput; (b) waste quality of each stream against BSFL requirements.

**Every animal has a "waste ceiling."** Feeding an animal too much waste-derived feed slows its growth — measurably. The model computes the exact ceiling for each species (quail: 41% of diet; BSFL: 71%, because it is a detritivore adapted to decaying material). The farm operates strictly below these ceilings, which is why contracted delivery volumes are reliable rather than hopeful.

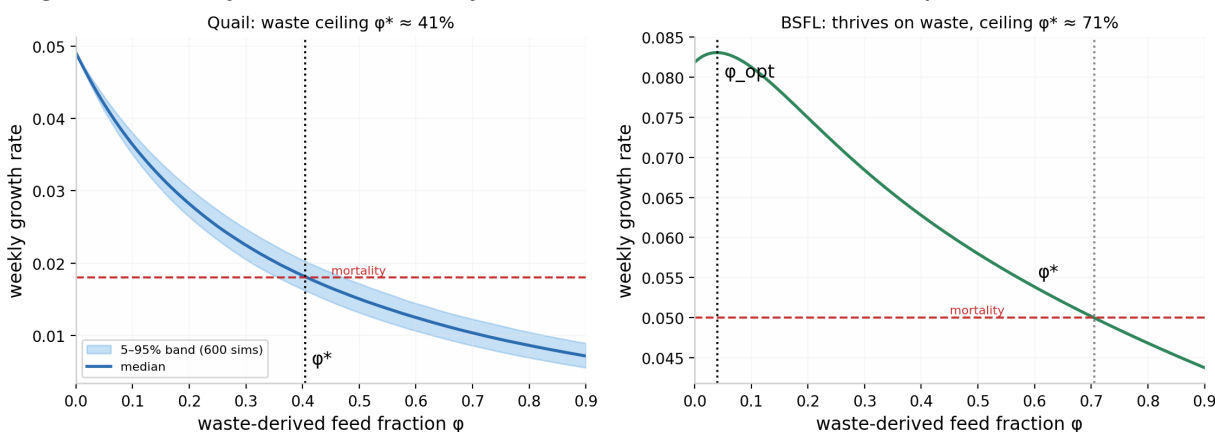


Figure 3 — growth rate versus waste-fraction of the diet. Quail: monotonic penalty, break-even at 41%. BSFL: peaks at a small waste fraction, tolerates up to 71%.

## 5. What the model decides for us each week

The system is run by explicit rules, not intuition. Four examples:

- **How many eggs to eat vs. hatch (the establishment trade).** Early on, the community deliberately eats fewer eggs so most can be incubated; the flock grows from a handful of birds to full capacity in about 20 weeks. Short-term scarcity is planned, not accidental.
- **How many birds can be processed (the culling governor).** A probabilistic rule computes the maximum number of birds that can be harvested this week while keeping the risk of falling below future breeding requirements under 1%. "Don't eat your breeding stock" is an equation.
- **What to sell and at what price (the channel allocator).** Every pound is routed to its highest-value buyer: human retail first, then wholesale, then pet food, then biomass — about \$922/week of export revenue from byproduct channels alone, re-allocating automatically when demand shifts (a simulated 50% retail-demand shock cost only ~6% of revenue).
- **When it is safe to grow.** No expansion step is taken unless nutrient balance, cash, and safety-stock constraints all stay green. Growth is a decision the model signs off on, not a hope.

Safety floors are hard constraints: minimum inventory of community food, seed reserves, and emergency stock are set aside **before** anything is exported. The system cannot "accidentally" sell the seed corn.

## 6. The numbers

| Metric                           | Value                                 | Basis                                                   |
|----------------------------------|---------------------------------------|---------------------------------------------------------|
| Initial investment (2-cell farm) | \$13,000 (grid) / \$37,000 (off-grid) | sourced equipment list; kit, structure, energy          |
| Steady-state revenue             | ~\$4,300 / month                      | product-allocation model, wholesale-conservative prices |
| Steady operating cost            | ~\$885 / month                        | feed, labor, energy, packaging                          |
| Steady net cash flow             | \$2,000 – \$2,700 / month             | full loop (quail + crickets + fish)                     |
| Cash break-even                  | month ~15 (grid) / ~26 (off-grid)     | 60–104 week simulations incl. ramp                      |
| Payback per expansion cell       | ~8–12 months                          | \$4,800 per additional production cell                  |
| Break-even robustness            | week 83 ± 1 across 40 stochastic runs | Monte-Carlo, full integrated simulation                 |

**Honesty box.** All prices, feed costs, and yields are conservative placeholders drawn from the literature and current market ranges. Every number improves (or is confirmed) by signed buyer commitments and local quotes. The structure of the model — the loop, the ceilings, the rules — does not depend on any single price being right.

## 7. Energy and structures

The full two-cell farm draws about 41 kWh per day, dominated by the incubator and brooder (about 19 kWh/day combined). The design answers this three ways:

- **Earth-sheltered structures.** Buildings are set halfway into the ground, using the soil as a thermal battery. This cuts total energy demand about 12% and stabilizes brooder temperatures year-round.
- **Phased solar with modular storage.** A 5 kW PV array plus 20 kWh of modular lithium batteries covers most days; a tri-fuel generator bridges extended deficits. Capacity expands by adding modules as revenue allows.
- **Solar concentrators, built on-site.** Concentrating collectors supply heat — warming brooder air and processing water — rather than electricity. Heat from concentration, power from photovoltaics.

Where grid power is available, it is used — cutting roughly \$19k of capex and pulling break-even toward month 15. Off-grid capability is the resilience story, not a requirement.

## 8. The business model

**Six revenue lines, diversified by design.** Eggs, quail meat, fish, crickets, BSFL protein, and greens/sprouts peak at different times and serve different buyers; if one line underperforms, the allocator shifts volume to the next channel automatically.

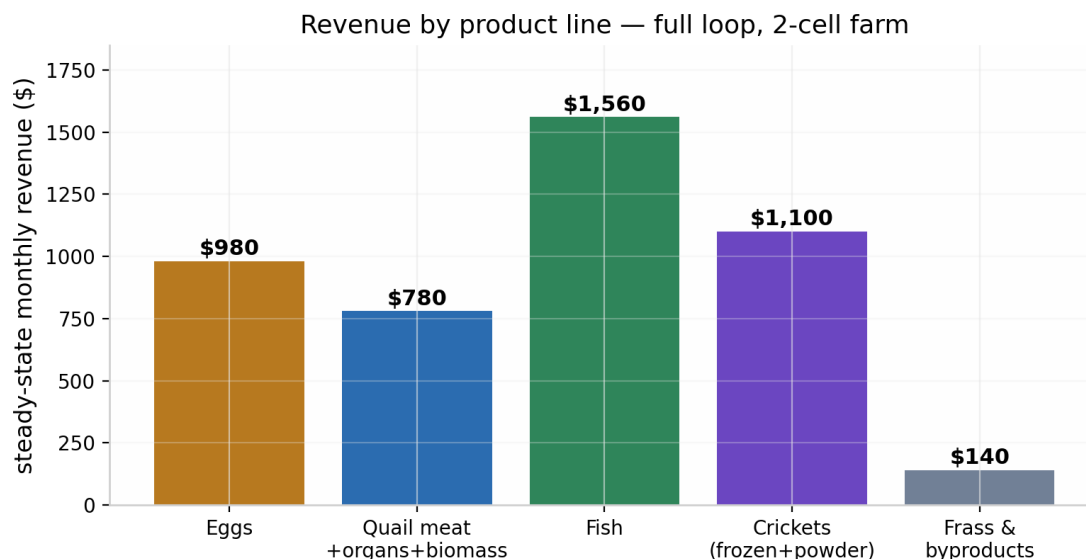


Figure 4 — steady-state monthly revenue by product line (2-cell farm, full loop).

**Buyers fund construction — food futures.** An anchor buyer (school, hospital, grocer, restaurant) signs a forward contract: scheduled deliveries at 10–15% below market, deposits escrowed, refundable if we fail to deliver. Pre-sales become zero-interest capital — no dilution, no debt service while building.

**Grants match buyers.** The project maps onto four standing USDA programs: Local Food Promotion Program and Value-Added Producer Grant (equipment and working capital), Beginning Farmer and Rancher Development Program (workforce training), and GusNIP (food-access incentives via our nonprofit partner). USDA grants reward exactly what we have: committed buyers, a feasibility model, and a cooperative structure.

## 9. The plan

| Phase                  | Months | What happens                                                                       | Gate to advance                           |
|------------------------|--------|------------------------------------------------------------------------------------|-------------------------------------------|
| <b>0 — Commit</b>      | 0–6    | sign 2–3 anchor LOIs; recruit founding members; secure site; feasibility study     | buyers + members signed                   |
| <b>1 — Build</b>       | 6–12   | earth-sheltered structure; 2 quail cells; phased energy; sprouts (fastest revenue) | futures deposits + grants secured         |
| <b>2 — Ramp</b>        | 12–24  | crickets and BSFL online; quail at full production; first contracted deliveries    | weekly delivery track record              |
| <b>3 — Steady loop</b> | 24+    | fish module; full six-line operation; surplus to food-access programs at cost      | all failure probabilities below threshold |

Each phase is gated by the model: no step is taken unless nutrient balance, cash, and safety-stock constraints all remain satisfied. The farm expands only when it is safe to expand.

## 10. Risks, and how the design answers them

| Risk                               | Mitigation (built into the model)                                                                  |
|------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Demand shortfall</b>            | futures LOIs signed before build; channel allocator re-routes automatically (shock-tested)         |
| <b>Production failure</b>          | species held below measured waste ceilings; staged ramp; weekly safety floors                      |
| <b>Price swings</b>                | sensitivity grid on every input; conservative wholesale base case                                  |
| <b>Energy interruption</b>         | grid-tie where available; phased solar + battery + tri-fuel backup; earth-sheltered load reduction |
| <b>Over-expansion</b>              | model-gated growth: all constraint probabilities must stay below threshold to advance              |
| <b>Key-person / knowledge risk</b> | everything is documented in executable notebooks — the farm is a manual, not a person              |

**What we are NOT claiming.** This is a pre-construction design, validated on paper and in simulation. The honest next step is small, instrumented, and reversible: build Phase 1, measure everything, and let the model learn.

## 11. How to be part of it

Four ways in — pick yours:

- **Invest.** \$150,000 funds the first farm (use of funds below). Instrument: a revenue-share note repaid from farm cash flow, or non-voting preferred member shares in the cooperative.
- **Buy the future.** Anchor buyers sign forward contracts — discounted, scheduled, local supply with a story your customers will love. Deposits are escrowed and refundable.
- **Live and work.** Founding resident roles include housing, training, a living wage, and ownership. Priority to people who are unemployed where the farm is built.
- **Advise.** Agronomy, entomology, aquaculture, food safety, law, fundraising — the model has placeholders with your name on them.

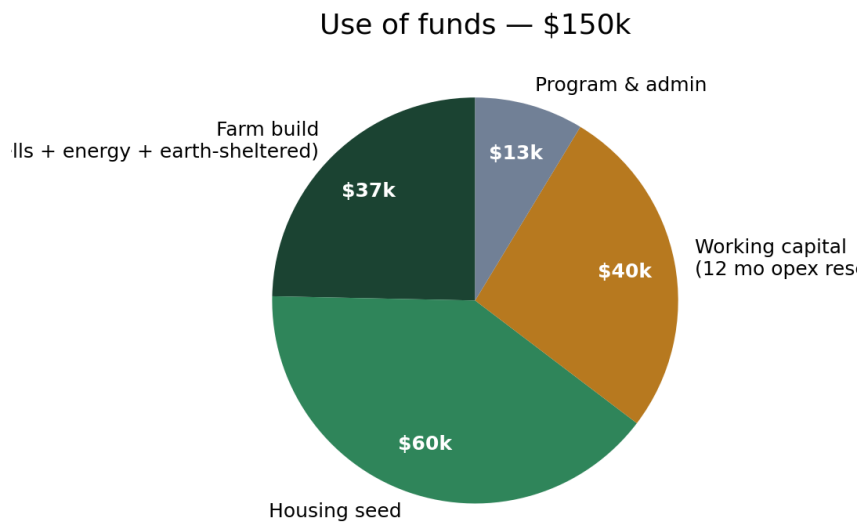


Figure 5 — use of funds for the \$150,000 raise.

**Contact:** [Your name] — [email] — [phone] — [site / signup link]

## Appendix — the technical artifacts

This report is the plain-language layer. Everything underneath is documented, sourced, and executable:

| Artifact                          | What it contains                                                                                                    | Format              |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------|
| Full mathematical model           | All eleven modules: demographics, growth, crops, energy, closure, stochastic control — equations plus runnable code | Jupyter notebook    |
| Research preprint                 | Peer-review-track paper: nutrient-flow network, waste quality index, causal effects, growth modulation              | arXiv (LaTeX + PDF) |
| Cost sheet v2                     | Sourced equipment capex, 60-month cash flows, energy audit, configuration comparisons                               | Excel workbook      |
| Product-allocation optimizer      | Linear program: channel tiers, demand caps, storage and safety floors, shadow prices                                | Excel + solved LP   |
| Futures LOI + grant concept paper | Buyer commitment instrument; VAPG scoring-mapped one-pager                                                          | Word documents      |
| Investor deck & community poster  | Presentation materials derived from this report                                                                     | PowerPoint          |

**Verification notes before public circulation:** confirm the USDA food-insecurity figure against the 2023 Household Food Security report; replace all bracketed placeholders; attach live buyer LOIs and local equipment quotes when available. All dollar figures are conservative planning values, not forecasts.